# Emergency Services Communication Platform

## Project Progress Recap

CE1 Group 3

# Recap

## Project goal

Provide a shared communication interface for emergency services.

The platform acts as a centralized communication layer between emergency-service users and systems.

# Architecture

**Before**

- Single-node Kubernetes cluster
- Hosted with Minikube

**Now**

- Multi-node Kubernetes cluster (k3s)
- Three worker nodes
- Managed with `kubectl`
- Portainer.io used for visualization

# Current Cluster Layout

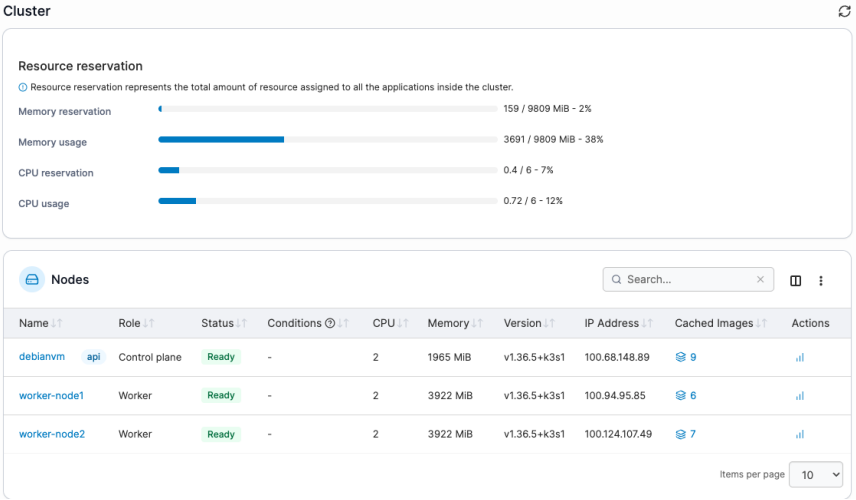
## Physical machine 1

Control plane + one worker node running in a virtual machine.

## Physical machine 2

Two worker nodes running in virtual machines.

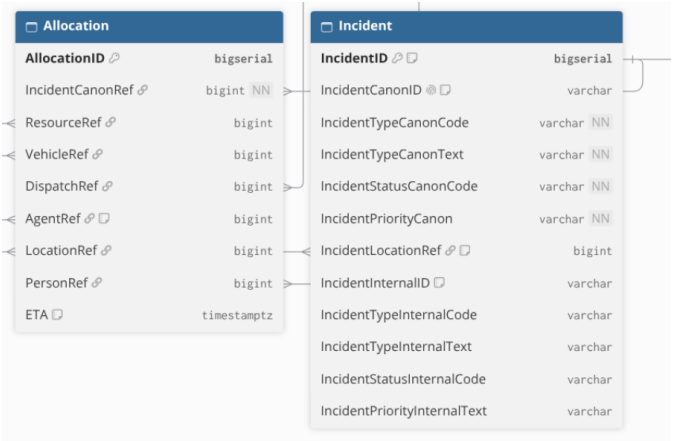
# Cluster Overview



# Data layer

**Data**

- Database schemas established.
- Basic database queries established.



# Domain

- Incident model established.
- Business rules for incidents established.
- Valid incident state transitions defined.

**Result**

Incident behavior is constrained by explicit domain rules rather than arbitrary status updates.

# Application

- API adapted to the incident domain.
- Basic CRUD operations are implemented.
- Seperation of concerns

- Transport $\leftrightarrow$ Application
- Application $\leftrightarrow$
  - Domain
  - Repository (ORM)

# Next Focus

1. Backend - API - WebSocket
2. Frontend - UI
3. Report